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BATCH 04

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING																		
Program Name: B. Tech		Assignment Type: Lab	Academic Year: 2025- 2026																	
Course Coordinator Name		Dr. Rishabh Mittal																		
Instructor(s) Name		<table border="1"> <tr><td>Mr. S Naresh Kumar</td></tr> <tr><td>Ms. B. Swathi</td></tr> <tr><td>Dr. Sasanko Shekhar Gantayat</td></tr> <tr><td>Mr. Md Sallauddin</td></tr> <tr><td>Dr. Mathivanan</td></tr> <tr><td>Mr. Y Srikanth</td></tr> <tr><td>Ms. N Shilpa</td></tr> <tr><td>Dr. Rishabh Mittal (Coordinator)</td></tr> <tr><td>Dr. R. Prashant Kumar</td></tr> <tr><td>Mr. Ankushavali MD</td></tr> <tr><td>Mr. B Viswanath</td></tr> <tr><td>Ms. Sujitha Reddy</td></tr> <tr><td>Ms. A. Anitha</td></tr> <tr><td>Ms. M.Madhuri</td></tr> <tr><td>Ms. Katherashala Swetha</td></tr> <tr><td>Ms. Velpula sumalatha</td></tr> <tr><td>Mr. Bingi Raju</td></tr> </table>		Mr. S Naresh Kumar	Ms. B. Swathi	Dr. Sasanko Shekhar Gantayat	Mr. Md Sallauddin	Dr. Mathivanan	Mr. Y Srikanth	Ms. N Shilpa	Dr. Rishabh Mittal (Coordinator)	Dr. R. Prashant Kumar	Mr. Ankushavali MD	Mr. B Viswanath	Ms. Sujitha Reddy	Ms. A. Anitha	Ms. M.Madhuri	Ms. Katherashala Swetha	Ms. Velpula sumalatha	Mr. Bingi Raju
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CourseCode	23CS002P C304	Course Title	AI Assisted Coding																	
Year/Sem	III/II	Regulation	R23																	
Date and Day of Assignment	Week1 – Wednesda y	Time(s)	23CSBTB01 To 23CSBTB52																	
Duration	2 Hours	Applicable to Batches	All batches																	
Assignment Number: 1.3(Present assignment number)/24(Total number of assignments)																				

Question	Expected Time to co

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1	Lab 2: Exploring Additional AI Coding Tools beyond Copilot – Gemini (Colab) and Cursor AI Lab Objectives: † To explore and evaluate the functionality of Google Gemini for Alassisted coding within Google Colab. † To understand and use Cursor AI for code generation, explanation, and refactoring. † To compare outputs and usability between Gemini, GitHub Copilot, and Cursor AI. † To perform code optimization and documentation using AI tools. Lab Outcomes (LOs): After completing this lab, students will be able to: † Generate Python code using Google Gemini in Google Colab. † Analyze the effectiveness of code explanations and suggestions by Gemini. † Set up and use Cursor AI for AI-powered coding assistance. † Evaluate and refactor code using Cursor AI features. † Compare AI tool behavior and code quality across different platforms.	Week 1 - Monday
	Task 1: Refactoring Odd/Even Logic (List Version) † Scenario: You are improving legacy code. † Task: Write a program to calculate the sum of odd and even numbers in a list, then refactor it using AI. † Expected Output: † Original and improved code † OUTPUT † PROMPT : generate python code Given a list of numbers, calculate the sum of even numbers and the sum of odd numbers separately. † CODE :	

	 <pre> numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10] even_sum = 0 odd_sum = 0 for num in numbers: if num % 2 == 0: even_sum += num else: odd_sum += num print(f"The sum of even numbers is: {even_sum}") print(f"The sum of odd numbers is: {odd_sum}") </pre> ✚ CODE OUPUT <pre> *** The sum of even numbers is: 30 The sum of odd numbers is: 25 </pre> EXPLANATION ✚ A list named <code>numbers</code> is created that contains integers from 1 to 10. ✚ Two variables, <code>even_sum</code> and <code>odd_sum</code>, are initialized to 0 to store the sums of even and odd numbers. ✚ A for loop goes through each number in the list. ✚ If a number is divisible by 2 (<code>num % 2 == 0</code>), it is even and added to <code>even_sum</code>; otherwise, it is odd and added to <code>odd_sum</code>. ✚ Finally, the program prints the sum of even numbers and the sum of odd numbers separately.	
	Task 2: Area Calculation Explanation ✚ Scenario: You are onboarding a junior developer. ✚ Task: Ask Gemini to explain a function that calculates the area of different shapes. ✚ Expected Output: ☒ Code ☐ Explanation ☐ OUTPUT ☐ PROMPT : Generate a python function that calculates the area of different shapes. ✚ CODE :	

```

import math

def calculate_area(shape_type, *args):
    shape_type = shape_type.lower()

    if shape_type == 'circle':
        if len(args) == 1:
            radius = args[0]
            return math.pi * radius**2
        else:
            return "Error: Circle requires one argument (radius)."
```

CODE OUTPUT:

```

# Test cases
print(f"Area of a circle with radius 5: {calculate_area('circle', 5)}")
print(f"Area of a square with side 4: {calculate_area('square', 4)}")
print(f"Area of a rectangle with length 6 and width 3: {calculate_area('rectangle', 6, 3)}")
print(f"Area of a triangle with base 10 and height 5: {calculate_area('triangle', 10, 5)}")

# Error cases
print(f"Unknown shape: {calculate_area('pentagon', 5)}")
print(f"Insufficient arguments for circle: {calculate_area('circle')}")
print(f"Too many arguments for square: {calculate_area('square', 4, 5)}")
```

Area of a circle with radius 5: 78.53981633974483
 Area of a square with side 4: 16
 Area of a rectangle with length 6 and width 3: 18
 Area of a triangle with base 10 and height 5: 25.0
 Unknown shape: Error: Unknown shape type. Supported shapes: circle, square, rectangle, triangle.
 Insufficient arguments for circle: Error: Circle requires one argument (radius).
 Too many arguments for square: Error: Square requires one argument (side).

EXPLANATION

The math module is imported to use the value of π (math.pi) for circle area calculation.

- ✚ A function calculate_area() is defined that takes a shape_type and a variable number of arguments using *args.
- ✚ The shape_type is converted to lowercase so the function works even if the user enters uppercase or mixed-case input.
- ✚ Based on the shape name (circle, square, rectangle, triangle), the function checks whether the correct number of arguments is provided.
- ✚ If the arguments are correct, it calculates and returns the area using the appropriate formula.
- ✚ If the arguments are incorrect or the shape is unknown, the function returns a clear error message explaining what is wrong.



Task 3: Prompt Sensitivity Experiment

Scenario:

You are testing how AI responds to different prompts.

Use Cursor AI with different prompts for the same problem and observe code changes.

Expected Output:

- Prompt list
- Code variations
- OUTPUT
- PROMPT : generate python code Given a list of numbers, calculate the sum of even numbers and the sum of odd numbers separately.(USING CURSOR AI)
- CODE

```
def calculate_sums(numbers):  
    """  
    Calculate the sum of even numbers and the sum of odd numbers separately.  
    Args:  
        numbers: A list of numbers  
    Returns:  
        A tuple containing (sum_of_evens, sum_of_odds)  
    """  
    sum_even = 0  
    sum_odd = 0  
  
    for num in numbers:  
        if num % 2 == 0:  
            sum_even += num  
        else:  
            sum_odd += num  
  
    return sum_even, sum_odd  
  
def main():  
    # Example usage  
    numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]  
  
    sum_even, sum_odd = calculate_sums(numbers)  
  
    print(f"List of numbers: {numbers}")  
    print(f"Sum of even numbers: {sum_even}")  
    print(f"Sum of odd numbers: {sum_odd}")  
  
if __name__ == "__main__":  
    main()
```

CODE OUTPUT :

```
$ python sum_even_odd.py  
List of numbers: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]  
Sum of even numbers: 30  
Sum of odd numbers: 25
```

- EXPLANATION :
- This program calculates the **sum of even numbers** and **sum of odd numbers** from a given list.

✚ Task:

✚

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† The list used is:

Feature	Gemini (Google Colab)	Cursor AI
Code Understanding	Clearly understands the logic and purpose of the program	Quickly understands the intent from exist code
Refactoring Style	Focuses on readability and simplicity	Focuses on optimization and concise synt
Suggested Improvement	Uses Pythonic constructs like generator expressions with <code>sum()</code>	Suggests optimized one-liners or function style code
Code Explanation	Provides detailed, beginner-friendly explanations	Provides short and technical explanations
Ease of Use	Easy to use via prompts in Google Colab	Integrated directly into the code editor
Output Accuracy	Produces correct and clean output	Produces correct and efficient output
Best For	Beginners and learning concepts	Developers focusing on speed and refactoring

[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

- † The program checks **each number** in the list:
 - If the number is divisible by 2, it is treated as **even** and added to the even sum.
 - Otherwise, it is treated as **odd** and added to the odd sum.
- † **Even numbers** in the list are: 2, 4, 6, 8, 10 Their sum is: **30**
- † **Odd numbers** in the list are: 1, 3, 5, 7, 9
Their sum is: **25**
 - Finally, the program prints the list along with the calculated sums of even and odd numbers as shown in the output.

Task 4: Tool Comparison Reflection

- † **Scenario:**
You must recommend an AI coding tool.
- † **Task:**
Based on your work in this topic, compare Gemini, Copilot, and Cursor AI for usability and code quality.
- † **Expected Output:** Short written reflection

OUTPUT PROMPT : COMPARING THE CODE (USING GEMINI AND CURSOR AI)

COMPARISON TABLE (USING GEMINI AND CURSOR AI)

	Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.	
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